

Deletion operation:

```
# Example:
if __name__ == "__main__":
    bst1 = BinarySearchTree()
    bst1.insert("AAAA", "555-1234")
    bst1.insert("BBBB", "555-2345")
    bst1.insert("CCCC", "555-3456")
    bst1.insert("DDDD", "555-5555")
    bst1.insert("EEEE", "555-5553")
    bst1.insert("CDSF", "555-5534")
    bst1.insert("ABCD", "555-5532")
    bst1.insert("ASED", "555-5643")
    bst1.insert("CSAD", "555-557")
    bst1.insert("ZZZZ", "555-555755")

    # Deleting a node
    bst1.delete("CCCC")
    bst1.delete("DDDD")
    bst1.inorder_traversal()
```

AAAA: 555-1234
ABCD: 555-5532
ASED: 555-5643
BBBB: 555-2345
CDSF: 555-5534
CSAD: 555-557
EEEE: 555-5553
ZZZZ: 555-555755

Searching Operation:

```
# Example:
if __name__ == "__main__":
    bst1 = BinarySearchTree()
    bst1.insert("AAAA", "555-1234")
    bst1.insert("BBBB", "555-2345")
    bst1.insert("CCCC", "555-3456")
    bst1.insert("DDDD", "555-5555")
    bst1.insert("EEEE", "555-5553")
    bst1.insert("CDSF", "555-5534")
    bst1.insert("ABCD", "555-5532")
    bst1.insert("ASED", "555-5643")
    bst1.insert("CSAD", "555-557")
    bst1.insert("ZZZZ", "555-555755")

    # Searching phone numbers
    print(bst1.search("AAAA"))
    print(bst1.search("D"))
    print(bst1.search("BBBB"))
```

555-1234

None

555-2345

Operation for insertion

```
# Example:
if __name__ == "__main__":
    bst1 = BinarySearchTree()
    bst1.insert("AAAA", "555-1234")
    bst1.insert("BBBB", "555-2345")
    bst1.insert("CCCC", "555-3456")
    bst1.insert("DDDD", "555-5555")
    bst1.insert("EEEE", "555-5553")
    bst1.insert("CDSF", "555-5534")
    bst1.insert("ABCD", "555-5532")
    bst1.insert("ASED", "555-5643")
    bst1.insert("CSAD", "555-557")
    bst1.insert("ZZZZ", "555-555755")

    bst1.inorder_traversal()

    # # Deleting a node
    # bst1.delete("CCCC")
    # bst1.inorder_traversal()
```

```
AAAA: 555-1234
ABCD: 555-5532
ASED: 555-5643
BBBB: 555-2345
CCCC: 555-3456
CDSF: 555-5534
CSAD: 555-557
DDDD: 555-5555
EEEE: 555-5553
ZZZZ: 555-555755
```